RESEARCH PROTOTYPE BRIEF

CetaSignal

A Constraint-Based Acoustic Classifier for Cetacean Coordination Signals

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Executive Summary

CetaSignal is a rule-based acoustic classification framework designed to infer whale behavioral coordination states from vocalization cadence features. The system is built on a constraint framework derived from ocean physics and cetacean ecology, which predicts that whale vocal signals evolved primarily to solve coordination problems in a low-visibility, three-dimensional marine environment.

Rather than treating whale communication as a symbolic language system, the CetaSignal model interprets acoustic signals as coordination protocols enabling group alignment and behavioral synchronization. The classifier maps measurable acoustic features to five behavioral coordination classes:

- **Contact** — pod cohesion
- **Dive** — descent synchronization
- **Forage** — prey detection coordination
- **Navigate** — migration heading alignment
- **Broadcast** — long-range signaling or advertisement

The system uses transparent decision rules derived from physical and biological constraints of the marine acoustic environment. Because the model is rule-based rather than a black-box machine learning approach, each classification decision can be traced directly to acoustic features and ecological reasoning.

The current research prototype evaluates approximately 5,000 acoustic specimens spanning 19 cetacean species across multiple ocean basins using publicly available hydrophone datasets and curated marine mammal acoustic archives. Acoustic features are extracted from recordings and behavioral classifications are generated without prior exposure to annotated behavioral outcomes, then compared against independently documented behavioral records.

The Constraint Framework

Cetacean acoustic communication evolved under the unique physical constraints of the ocean environment. Several fundamental characteristics of the marine medium shape how information can be transmitted between individuals.

Low Visibility

In most ocean environments, visual range is severely limited. Coastal waters often provide visibility of only a few meters, while even the clearest open-ocean conditions rarely exceed tens of meters. Because cetacean pods may operate at spatial separations greater than these distances, visual communication is insufficient for maintaining group cohesion.

Three-Dimensional Spatial Coordination

Unlike terrestrial animals operating primarily within a two-dimensional plane, cetaceans must coordinate movement in a fully three-dimensional environment. Depth changes, heading adjustments, and lateral spacing occur simultaneously. Signals that coordinate behavior must therefore encode directional and temporal information relevant to this volumetric environment.

Acoustic Propagation in Water

Sound propagates efficiently through seawater, especially at low frequencies. In particular, the SOFAR (Sound Fixing and Ranging) channel acts as a natural acoustic waveguide that can allow low-frequency whale calls to travel across basin-scale distances with relatively low attenuation.

This physical property creates a strong evolutionary incentive for acoustic signaling systems optimized for:

- Long-range propagation
- Reliable detection in noisy environments
- Rapid coordination across spatially distributed individuals

Frequency-Dependent Attenuation

Higher frequency sounds attenuate much more rapidly in seawater than low-frequency signals. As a result, different signal frequencies naturally correspond to different effective communication ranges. Low-frequency calls are suited for long-distance navigation or broadcast signaling, while higher frequency signals provide greater spatial resolution for short-range coordination or echolocation.

Resulting Hypothesis

These constraints suggest that whale vocalizations evolved not primarily as symbolic language units but as behavioral coordination signals shaped by the physical environment. Vocal signals serve to reduce uncertainty among distributed individuals by communicating behavioral state changes or proposed actions.

Examples of coordination problems include:

- Maintaining pod cohesion
- Initiating dives
- Coordinating foraging maneuvers
- Synchronizing migration heading
- Advertising presence across large distances

Acoustic Feature Extraction

The classifier relies on acoustic features that can be measured directly from recorded signals.

Peak Frequency

Represents the dominant frequency of the signal within a given time window, measured using spectral analysis techniques such as the Fast Fourier Transform (FFT) applied to a spectrogram. Corresponds to the frequency bin with the highest power.

Frequency Contour

Describes how the dominant frequency of a signal changes over time, determined by tracking the peak frequency across successive spectrogram frames. Categorized as rising, descending, flat, or complex (multiple inflections). Often reflects directional movement or behavioral state changes.

Call Duration

Measured as the time difference between the detected start and end of a vocalization. Signal boundaries are determined using amplitude thresholds or energy-based detection methods.

Inter-Pulse Interval (IPI)

For pulsed signals, represents the time spacing between successive pulses. Especially relevant for odontocete click trains and social codas.

Repetition Rate

Measures the frequency with which calls occur within a given time window. Expressed as calls per second or calls per minute.

Urgency Index

Approximates the temporal density of information within a signal. Formulated as: *Urgency Index = (Pulse Density × Repetition Rate) / Signal Duration*. Higher values correspond to signals produced during rapid behavioral coordination such as prey pursuit.

Behavioral Coordination Classes

The CetaSignal classifier maps acoustic features to five coordination categories.

Contact

Signals that maintain pod cohesion when individuals are spatially separated. Often exhibit rising frequency contours and moderate durations. Associated with convergence behaviors between individuals.

Dive

Signals that coordinate the initiation of descent. Frequently display descending frequency contours and short durations, consistent with rapid behavioral transitions prior to submergence.

Forage

Signals occurring during prey detection or hunting coordination. Often consist of high-frequency pulses or burst-like click trains with elevated urgency indices.

Navigate

Signals associated with sustained directional movement such as migration. Tend to occupy low frequency ranges and repeat at slow intervals over extended periods.

Broadcast

Long-range signals not tied to immediate group coordination. Examples include humpback whale songs or complex vocal displays associated with reproductive signaling or presence advertisement.

Example Case Study

An example from North Atlantic right whale recordings illustrates the classification approach.

Feature	Measurement
Species	North Atlantic Right Whale
Signal Type	Upcall
Peak Frequency	~170 Hz
Frequency Contour	Rising
Call Duration	~1.2 seconds
Inter-Pulse Interval	None detected
CetaSignal Classification	**Contact**

Independent behavioral observations associated with the recording indicate that two whales initially separated by approximately 200 meters converged to within roughly 50 meters following the exchange of upcalls — consistent with a Contact/cohesion signal classification.

Dataset and Evaluation

The research prototype currently evaluates approximately 5,000 acoustic specimens drawn from publicly available marine bioacoustic datasets and curated hydrophone archives, spanning multiple cetacean species across several ocean basins.

Each specimen includes:

- A recorded acoustic signal
- Extracted acoustic features
- An associated behavioral annotation derived from observational records

During evaluation, the classifier receives only acoustic features and produces a behavioral classification. The predicted class is compared against the independently documented behavioral record.

Performance evaluation metrics include:

- Overall classification accuracy
- Class-specific recall and precision
- Confusion matrices across behavioral classes

Because the model is rule-based and interpretable, classification outcomes can be traced to individual acoustic features and decision rules.

Future Research Directions

Integration with Tag Data

Combining acoustic recordings with high-resolution movement data from biologging devices such as DTAGs could provide stronger validation by directly correlating vocal signals with movement behaviors.

Playback Experiments

Controlled playback experiments could test whether specific acoustic signal structures reliably trigger predicted behavioral responses.

Cross-Species Analysis

Comparative analysis across cetacean species could determine whether coordination signal structures are conserved or species-specific.

Juvenile Signal Acquisition

Understanding how juvenile whales acquire and refine vocal signals during development may provide insight into how coordination signals are learned and culturally transmitted.

Conclusion

CetaSignal represents an exploratory research prototype investigating how acoustic signal structure relates to behavioral coordination in cetaceans. The framework emphasizes interpretable classification grounded in physical and ecological constraints rather than opaque machine learning models.

By focusing on measurable acoustic features and coordination functions, the system provides a structured approach for analyzing whale vocalizations across species and environments. Further empirical validation and collaboration with marine bioacoustics researchers may help refine the model and identify experimental approaches capable of testing its predictions.